

1. Abstract

Vietnam records thousands of occupational accidents annually, with PPE non-compliance and safety violations among the leading causes, many linked to PPE non-compliance. Cloud-based monitoring suffers high latency, bandwidth cost, and privacy risk. This project deploys a lightweight YOLO26n edge-AI pipeline for real-time PPE and hazard-zone detection: **131 FPS (ONNX FP32)**, **94.6% fewer false alarms** via **5-frame temporal smoothing**, and **99.93% bandwidth savings (1.35 MB/hour)** — a scalable, low-latency, privacy-preserving safety solution.

2. Motivation & Objective

Vietnam recorded: **7,004 accidents, 658 deaths** (2025) — **13.75% procedure violations, 6.25% missing PPE** [1]. CCTV monitoring is fatigue-prone; cloud AI adds latency (**500–1000 ms**), **bandwidth (4–8 Mbps/cam)**, **privacy risk**. Our objective:

- > Detect PPE violations & hazard zones in real time
- > Reduce false alarms via temporal smoothing
- > Minimize bandwidth with edge-first design

3. Methodology

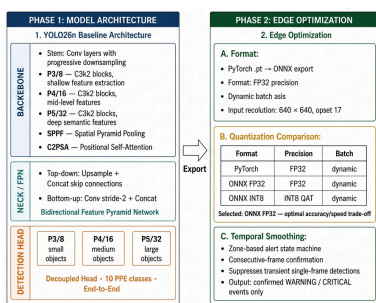


Fig 2: Overall method

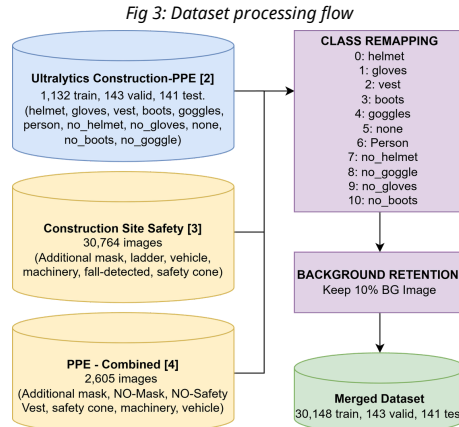


Fig 3: Dataset processing flow



Fig 1: PPE violated construction site

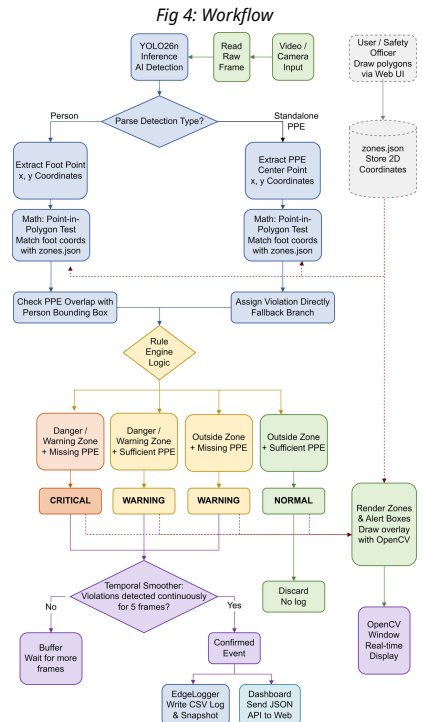


Table 2: ONNX Quantization Results

4. Results

Table 1: Edge Optimization Results

(*Assume 1 event/min).

Optimization Target	Baseline State	Edge-Optimized Result
Alert Reliability (baseline_yolo26n ONNX FP32)	124 alerts (No temporal filter)	6 alerts (N = 5 frames)
Network Efficiency	1.897 GB/hr (Raw video)	1.35 MB/hr (Logs + Snapshots)*

Model Format	Model Size	Latency (ms/img)	Throughput (FPS)	Overall mAP@0.5	Violation mAP@0.5
ONNX FP32 (Dynamic B=8)	10.3 MB	761	131	749	671
ONNX FP16 (Dynamic B=4)	5.7 MB	905	110	750	671

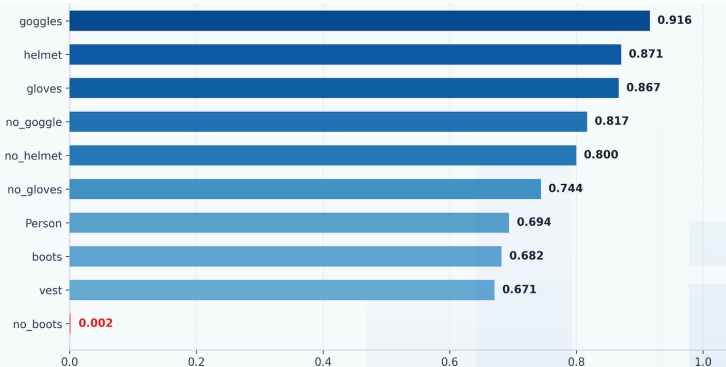


Figure 5: mAP@0.5 result



Figure 6: Snapshot of a violated scene

5. Conclusion

- Successfully deployed an edge AI pipeline for real-time PPE monitoring, eliminating cloud latency, bandwidth, and privacy issues.
- The optimized YOLO26n model (ONNX FP32) achieves 131 FPS (2.93 ms/frame), enabling multi-camera processing.
- A 5-frame temporal smoothing algorithm reduces false alarms by 94.6%.
- The edge-first architecture reduces bandwidth usage by 99.93%, requiring only 1.35 MB/hour.
- **Future work:** improve no-boots detection, integrate BIM-based dynamic safety zones, and apply multi-modal sensor fusion for adverse weather conditions.

6. References

- [1] M. Hien, "Tình hình tai nạn lao động năm 2025: Giảm về số vụ nhưng vẫn tiềm ẩn nhiều rủi ro," Tạp chí Tổ chức nhà nước và Lao động - Bộ Nội vụ, Mar. 2026.
- [2] M. Dalvi et al., "Construction-PPE: Personal Protective Equipment Detection Dataset," Ultralytics, Jan. 2025.
- [3] Roboflow Universe Projects, "Personal Protective Equipment - Combined Model Dataset," Roboflow Universe, Jun. 2025.
- [4] Roboflow Universe Projects, "Construction Site Safety Dataset," Roboflow Universe, Jan. 2026.